

A polynomial rate for the uniform endpoint profile

Complete asymptotic proof; independently reviewed

30 September 2026

Assume $p_1, \dots, p_m : [K] \rightarrow [0, 1]$ are nondecreasing, with positive probability row weights w , and

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

Put $u_i = 1 - p_i$ and $s = \sum_i u_i$.

Theorem 1. *For every fixed $0 < A < 1$, there is a constant C_A such that, for all sufficiently large m , uniformly over columns and rows with $s \leq A \log m$,*

$$\max_i |m(1 - p_i) - s| \leq C_A m^{-(1-A)/8} \log m. \quad (1)$$

In particular this is a quantitative bound on the range of the comparison columns, rather than only a bound on their maximum. No new dice-face exponent is asserted in this note.

We use the already reviewed marked comparison from `quantitative_relative_endpoint_balance.tex`. For a fixed column i , put $M = m - 1$, $\ell = \lfloor M/4 \rfloor$, $\alpha = \ell/M$, $s_- = \sum_{j \neq i} u_j$, and $x = \alpha s_-$. For a uniform ℓ -subset B of the remaining columns let $\bar{A} = \mathbb{E}_B \prod_{j \in B} p_j$. The two probability measures

$$\mu_0 = (\ell + 1)w\bar{A}, \quad \mu_1 = (\ell + 1)(\ell + 2)wu_i\bar{A}$$

have density relation $d\mu_1 = (\ell + 2)u_i d\mu_0$. Their scalar targets have densities

$$b_0(z) = \frac{\ell + 1}{\ell} (1 - z/\ell)^\ell \mathbf{1}_{[0, \ell]}, \quad b_1(z) = \frac{\ell + 2}{\ell} z b_0(z). \quad (2)$$

Denote their integrals by β_0, β_1 , and fix $\theta = 1/16$. All constants in the auxiliary bounds below are uniform in i .

Lemma 1 (The degree range in the marked comparison). *For some absolute $c_0 > 0$, the marked bilinear comparison holds for all degrees $D \leq c_0 M$:*

$$|\mu_a(Q_1 Q_2) - \beta_a(Q_1 Q_2)| \leq C B_1 B_2 (D/M + e^{-cM}), \quad a = 0, 1, \quad (3)$$

where $Q_b = \sum_{j=0}^D c_{bj} L_j(\theta z)$ and $B_b = \sum_j |c_{bj}|$. Also

$$\mu_a(e^{\theta x/2}) + \beta_a(e^{\theta z/2}) \leq C. \quad (4)$$

Proof. In the proof of the cited comparison, the polarization series is bounded by a constant times

$$B_1 B_2 \sum_{k=2}^{2D} (C \sqrt{D/M})^k (k+1)^C.$$

The exponent on $k+1$ is fixed, also for the marked measure. Taking c_0 sufficiently small makes the geometric ratio at most $1/4$, so this remains $O(B_1 B_2 D/M)$. Choose also $2c_0 M \leq M - \ell$. The weighted-subset comparison and its exponential-tail argument have no smaller degree restriction; $D \leq c_0 M$ is polynomially bounded and its tail remains exponentially small. This proves (3). Bound (4) follows from $\bar{A} \leq e^{-\alpha s_-}$, the row-mass Laplace bound, and $u_i \leq C(s_- + 1)/M$ for the marked normalization. It is also immediate for the scalar densities (2). $\square$

An elementary Laguerre smoothing kernel

Let $0 < r < 1$, $\tau = 1 - r$. Given $X = x \geq 0$, choose

$$N \sim \text{Poisson}(r\theta x/\tau), \quad Y \mid N \sim (\tau/\theta) \text{Gamma}(N+1, 1). \quad (5)$$

Write $T_r f(x) = \mathbb{E}[f(Y) \mid X = x]$. The following standard Laguerre kernel facts are included with proofs. The associated classical bilinear formula is the Hille–Hardy formula; see the primary reference [1, 18.18.27].

Lemma 2. *The kernel T_r is stochastically increasing in x , reversible for $\rho_\theta(dx) = \theta e^{-\theta x} dx$, and satisfies*

$$T_r L_j(\theta x) = r^j L_j(\theta x).$$

If f is an indicator function and $a_j = \int f(x) L_j(\theta x) d\rho_\theta(x)$, then $\sum_j a_j^2 \leq 1$ and

$$T_r f(x) = \sum_{j=0}^{\infty} r^j a_j L_j(\theta x). \quad (6)$$

After truncation at degree D , the remainder is at most

$$C\tau^{-1/2}r^{D+1}e^{\theta x/2}. \quad (7)$$

Proof. Increasing x can be coupled by adding an independent Poisson variable to N , then adding the corresponding independent unit-rate exponentials to the gamma sum. This proves stochastic monotonicity. Summing the gamma densities gives the transition density

$$\frac{\theta}{\tau} e^{-(\theta y + r\theta x)/\tau} \sum_{n=0}^{\infty} \frac{(r\theta^2 xy/\tau^2)^n}{(n!)^2}.$$

Multiplication by $\theta e^{-\theta x}$ makes this symmetric in x, y , proving reversibility and stationarity. The conditional moment generating function is

$$\mathbb{E}(e^{tY} \mid x) = (1 - \tau t/\theta)^{-1} \exp\left(\frac{rxt}{1 - \tau t/\theta}\right). \quad (8)$$

Applying this to the ordinary Laguerre generating function $(1-z)^{-1} \exp[-\theta xz/(1-z)]$ gives the same generating function with z replaced by rz , proving the eigenfunction identity. The Laguerre polynomials form an orthonormal basis of $L^2(\rho_\theta)$. Thus reversibility and the eigenfunction identity prove (6) in L^2 , with Bessel's inequality giving the coefficient bound. The series converges absolutely and uniformly on compact intervals, since $|L_j(\theta x)| \leq e^{\theta x/2}$; the transition integral is continuous for an interval indicator, and hence agrees pointwise with that series. Alternatively the displayed transition kernel and its Laguerre expansion give the same identity directly. Cauchy-Schwarz bounds the tail by $e^{\theta x/2}(\sum_{j>D} r^{2j})^{1/2}$, proving (7). Only interval indicators are needed below. $\square$

Lemma 3 (Uniform crossing concentration). *There are absolute constants $C, c > 0$ such that, if $L \geq 1$, $0 < h \leq 1$, $h \geq C\tau(L+1)$, and $0 \leq x \leq L+1$, then*

$$\Pr(|Y - x| > h \mid x) \leq 2 \exp[-ch^2/(\tau(L+1))]. \quad (9)$$

Proof. Formula (8) gives $\mathbb{E}(Y \mid x) = rx + \tau/\theta$. For $|t|\tau/\theta \leq 1/2$, subtracting this mean in its logarithm gives

$$\log \mathbb{E}[e^{t(Y - \mathbb{E}Y)} \mid x] \leq C\tau(x + \tau)t^2.$$

Indeed $-\log(1-v) - v \leq Cv^2$ and $t/(1-v) - t = tv/(1-v)$, with $v = \tau t/\theta$. The deterministic mean displacement has magnitude at most $C\tau(L+1)$. A Chernoff bound with $|t| = ch/(\tau(L+1))$, after increasing the constant in the hypothesis on h , proves (9). $\square$

Distribution functions with a polynomial error

Lemma 4. *For $a = 0, 1$ and $1 \leq L \leq \log M$, uniformly in $0 \leq z \leq L$,*

$$|\mu_a(x \leq z) - \beta_a([0, z])| \leq \varepsilon_L := CM^{-1/4} \sqrt{(L+1) \log M}. \quad (10)$$

Proof. Take $\tau = M^{-1/2}$ and truncate (6) at $D = \lfloor c_0 M \rfloor$. Applying (3) to the individual products $1 \cdot L_j$, summing, and using Cauchy–Schwarz gives, for any interval indicator f ,

$$\begin{aligned} |\mu_a(T_r f) - \beta_a(T_r f)| &\leq \frac{C}{M} \sum_{j \leq D} j r^j |a_j| + C \tau^{-1/2} e^{-c\tau M} \\ &\leq \frac{C}{M \tau^{3/2}} + C \tau^{-1/2} e^{-c\tau M} \leq CM^{-1/4}. \end{aligned}$$

The tails use (4) and (7); the constant polynomial has zero comparison error. The series $\sum j^2 r^{2j}$ is $O(\tau^{-3})$.

Choose $h = C_1 \sqrt{\tau(L+2) \log M}$ with C_1 sufficiently large. For large M , it is at most $1/4$ and satisfies the hypothesis of Lemma 3. The crossing error is at most M^{-10} . For each threshold $z \in [0, L]$, stochastic monotonicity and (9) therefore imply

$$(1 - M^{-10}) \mathbf{1}_{x \leq z} \leq T_r \mathbf{1}_{[0, z+h]}(x) \leq \mathbf{1}_{x \leq z+2h} + M^{-10}.$$

For x outside the concentration interval, stochastic monotonicity reduces the bound to $x = z$ or $x = z + 2h$; thus these inequalities hold on the whole half-line. The reverse comparison follows by shifting the threshold down by h , with a trivial lower bound when $z < 2h$. Integrate these inequalities against μ_a and β_a , use the smoothed-test comparison above, and note that both scalar densities in (2) are bounded by an absolute constant. This proves (10). It also handles atoms of μ_a and either choice of strict endpoints. $\square$

Recovering the monotone density

Lemma 5. *Suppose $1 \leq L \leq \log M$ and*

$$h_* = \sqrt{(L+1) \varepsilon_{L+2} e^L} \longrightarrow 0$$

along the range under consideration. Uniformly over rows with $x \leq L$,

$$|\ell u_i - x| \leq C h_*. \quad (11)$$

Here (10) can be used with $L+2$; replacing the upper limit $\log M$ there by $\log M + 2$ only changes constants.

Proof. For $0 \leq z \leq L+2$, the explicit scalar density satisfies $b_0(z) \geq ce^{-z}$ for large M , because $L = O(\log M)$ and $\ell \log(1 - z/\ell) = -z + O((\log M)^2/M)$. Choose an interval of length h_* lying strictly above the target row's value x , at distance between h_* and $2h_*$. Every row in it lies before the target row, so monotonicity and the exact density relation give

$$(\ell + 2)u_i(q) \leq \frac{\mu_1(I)}{\mu_0(I)}.$$

The scalar denominator is at least ch_*e^{-L} , and the two interval masses have errors at most $2\varepsilon_{L+2}$ by (10). For large M the denominator error is less than half the scalar mass, since $\varepsilon_{L+2}e^L/h_* = h_*/(L+1) \rightarrow 0$. The scalar density ratio $b_1/b_0 = ((\ell+2)/\ell)z$ varies by $O(h_*)$ across the chosen interval. Perturbing numerator and denominator therefore bounds its ratio error by

$$C(L+1)\varepsilon_{L+2}e^L/h_* = Ch_*.$$

This proves the upper bound in (11). For $x \geq 2h_*$ use the corresponding interval strictly below x for the lower bound. When $x < 2h_*$, nonnegativity supplies that bound directly. Flat row blocks cause no problem because the comparison intervals are strictly separated from the target value. $\square$

Proof of Theorem 1. For a row with $s \leq A \log m$ one has $x = \alpha s_- \leq (A/4) \log m$. Set $L = \max(1, (A/4) \log m)$. Then

$$h_* \leq C_A M^{-1/8} e^{L/2} (L+1)^{3/4} (\log M)^{1/4} \leq C_A m^{-(1-A)/8} \log m \rightarrow 0.$$

Lemma 5 applies uniformly in the deleted index i . Finally the conversion to the full mean is exact:

$$\frac{M}{\ell}(\ell u_i - x) = M u_i - s_- = m u_i - s.$$

Since M/ℓ is bounded, this gives (1). $\square$

References

- [1] NIST Digital Library of Mathematical Functions, *Hille–Hardy formula*, equation 18.18.27. <https://dlmf.nist.gov/18.18.E27>.